

IMAT MOCK EXAM 1 — "Baseline"

60 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · personal target: 47/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "Vaccination operates on the principle of immunological memory. Upon first exposure to an antigen, naïve lymphocytes proliferate and differentiate, generating both effector cells that clear the pathogen and long-lived memory cells. A subsequent encounter with the same antigen triggers a faster and stronger response because memory cells respond at lower antigen concentrations and divide more rapidly than naïve cells."

Which of the following is not supported by the passage?

- A) The secondary immune response exceeds the primary one in magnitude.
- B) Effector cells contribute to pathogen clearance.
- C) Naïve lymphocytes proliferate during the primary response.
- D) Memory cells persist after the initial infection has been cleared.
- E) Memory cells require a higher antigen concentration to activate than naïve cells.

Answer: E

The passage states that memory cells respond at LOWER antigen concentrations than naïve cells, so the claim that they require a higher concentration to activate says the opposite of the text and is not supported. The other four statements all follow from the passage: long-lived memory cells persist after the pathogen is cleared, the secondary response is faster and stronger (so larger in magnitude), naïve lymphocytes proliferate on first exposure, and effector cells clear the pathogen. The tempting trap is the magnitude claim, because it looks like an added detail, yet the passage's 'faster and stronger response' explicitly backs it.

2. "Enzymes lower the activation energy of a reaction without altering the free energy difference between substrates and products. Consequently, an enzyme changes the rate at which equilibrium is reached but never the position of equilibrium."

Which statement is a valid inference?

- A) The ratio of products to substrates at equilibrium is unchanged by the enzyme.
- B) Adding an enzyme increases the equilibrium constant.
- C) Enzymes shift the equilibrium toward products.
- D) A reaction that is unfavourable thermodynamically can be made favourable by adding more enzyme.
- E) An enzyme can make an endergonic reaction exergonic.

Answer: A

An enzyme lowers the activation energy but leaves the free energy difference between substrates and products unchanged, so the equilibrium position, and therefore the products-to-substrates ratio at equilibrium, is exactly the same with or without enzyme. Every claim that the enzyme makes an endergonic reaction exergonic, raises the equilibrium constant, shifts equilibrium toward products, or turns an unfavourable reaction favourable would require changing the free energy difference, which the passage rules out. More enzyme only means equilibrium is reached sooner, never a different equilibrium.

3. (General knowledge) Which of the following Italian scientists is best known for discovering a method of staining nervous tissue that allowed the visualisation of individual neurons, a contribution that shared the 1906 Nobel Prize in Physiology or Medicine?

- A) Antonio Meucci
- B) Rita Levi-Montalcini
- C) Galileo Galilei
- D) Camillo Golgi
- E) Guglielmo Marconi

Answer: D

Camillo Golgi developed the silver-based 'black reaction' for staining nervous tissue, which first allowed individual whole neurons to be seen under the microscope, and he shared the 1906 Nobel Prize in Physiology or Medicine with Ramon y Cajal. The decisive facts are the staining method plus the 1906 prize. The other names belong to different fields: Galileo (astronomy and physics), Levi-Montalcini (nerve growth factor, 1986 Nobel), Meucci (early telephone work), Marconi (radio, 1909 Nobel in Physics).

4. Five friends — A, B, C, D, E — sit in a row. B sits immediately to the right of A. C is not adjacent to A. D sits at one of the ends. E sits to the left of D. Which of the following must be true?

- A) E is not at an end.
- B) D sits at the far right.
- C) B is to the left of C.
- D) A is at the far left.
- E) C sits between B and E.

Answer: B

Givens: B immediately right of A; C not adjacent to A; D at one of the two ends; E somewhere to the left of D.

Deduction: D must occupy an end, so D sits far left or far right.

Deduction: E requires a seat strictly to the left of D.

Deduction: if D sat far left, no seat would exist to its left for E, contradicting the givens.

Conclusion: D is forced to the far right in every valid arrangement.

Tempting claim: that A sits at the far left fails because the pair A, B can start one seat in, as in the valid seating E, A, B, C, D.

5. All patients with disease X have elevated marker M. Some patients with disease Y also have elevated marker M. No patient with disease X has disease Y. From these, which conclusion necessarily follows?

- A) No patient with disease Y has elevated M.
- B) Every patient with elevated M has disease X or Y.
- C) Some patients with disease Y do not have disease X.
- D) Anyone with elevated M has disease X.
- E) Patients with disease X always have disease Y.

Answer: C

Givens: every disease X patient has elevated marker M; at least one disease Y patient has elevated M; no patient has both X and Y.

Deduction: there exists at least one patient with disease Y and elevated M (stated directly).

Deduction: that patient cannot have disease X, because X and Y never occur in the same patient.

Conclusion: some patients with disease Y do not have disease X, so this follows necessarily.

Tempting claim: that everyone with elevated M has X or Y fails because people with neither disease could also show elevated M; the givens never say M is exclusive to X and Y.

6. In a certain code, READING is written as SFB EJ OH. How is MONKEY written?

- A) NPO LEZ B) NPO MFZ C) NPO LFY D) NPO LFZ E) NQO LFZ

Answer: D

Givens: READING is encoded as SFB EJ OH; compare position by position.

Deduction: R to S, E to F, A to B, D to E, I to J, N to O: every letter moves one place forward in the alphabet, and the break after three letters is kept.

Apply the same shift to MONKEY: M to N, O to P, N to O, K to L, E to F, Y to Z.

Conclusion: MONKEY becomes NPO LFZ.

Tempting claim: a variant like NPO MFZ fails at the fourth letter, because K shifted one forward is L, not M; check every position, not just the first few.

7. A clock shows 3:15. What is the angle between the hour and minute hands?

- A) 0° B) 7.5° C) 15° D) 22.5° E) 30°

Answer: B

Given: the clock reads 3:15; the minute hand sweeps 6° per minute and the hour hand 30° per hour, that is 0.5° per minute.

Relationship: minute hand angle from 12 = minutes $\times 6^\circ$; hour hand angle = hours $\times 30^\circ$ + minutes $\times 0.5^\circ$; the answer is the difference of the two.

Substitute: minute hand = $15 \times 6^\circ = 90^\circ$; hour hand = $3 \times 30^\circ + 15 \times 0.5^\circ = 90^\circ + 7.5^\circ = 97.5^\circ$.

Hop: $97.5^\circ - 90^\circ = 7.5^\circ$

Result: the angle between the hands is 7.5° .

Classic slip: answering 0° assumes the hour hand sits exactly on the 3 mark at 3:15, but it has already crept 7.5° past it.

8. If the day after tomorrow is Wednesday, what day was it three days before yesterday?

- A) Saturday B) Tuesday C) Friday D) Sunday E) Thursday

Answer: E

Givens: the day after tomorrow is Wednesday.

Deduction: today is two days before Wednesday, so today is Monday.

Deduction: yesterday was therefore Sunday.

Deduction: counting back one day at a time from Sunday gives Saturday, then Friday, then Thursday.

Conclusion: three days before yesterday was Thursday.

Tempting claim: Friday fails by one day, from counting the three days back from today (Monday) instead of from yesterday (Sunday).

9. A sequence begins 2, 6, 12, 20, 30, ... What is the next term?

- A) 36 B) 40 C) 42 D) 44 E) 50

Answer: C

Given: the sequence 2, 6, 12, 20, 30, ...

Relationship: take successive differences to expose the growth pattern.

Hops: $6 - 2 = 4$; $12 - 6 = 6$; $20 - 12 = 8$; $30 - 20 = 10$

Deduction: each difference grows by 2, so the next difference is $10 + 2 = 12$.

Cross-check: the terms are $n \times (n + 1)$, so the sixth term is $6 \times 7 = 42$.

Result: $30 + 12 = 42$

Classic slip: 40 reuses the last difference of 10 instead of extending the growing gaps.

Section B - Biology (Q10-32)

10. In which cellular compartment does the Krebs (citric acid) cycle take place?

- A) Mitochondrial matrix
- B) Rough endoplasmic reticulum
- C) Mitochondrial intermembrane space
- D) Nucleus
- E) Cytosol

Answer: A

The Krebs cycle runs in the mitochondrial matrix, where pyruvate is imported, decarboxylated to acetyl-CoA by pyruvate dehydrogenase, and progressively oxidised. Glycolysis is the cytosol stage, and the intermembrane space is where protons accumulate for chemiosmosis, not where the cycle turns.

11. How many net ATP molecules are produced by substrate-level phosphorylation during glycolysis per molecule of glucose?

- A) 0 B) 1 C) 2 D) 4 E) 6

Answer: C

Glycolysis spends 2 ATP in the investment phase (hexokinase and phosphofructokinase) and makes 4 ATP in the payoff phase (phosphoglycerate kinase and pyruvate kinase), both by substrate-level phosphorylation, giving a net of $4 - 2 = 2$ ATP per glucose. The trap is quoting the gross yield of 4 and forgetting the 2 invested during the early steps.

12. Which molecule serves as the final electron acceptor of the mitochondrial electron transport chain?

- A) Water B) Pyruvate C) FAD D) Oxygen E) NAD^+

Answer: D

Oxygen is the terminal electron acceptor: at complex IV it accepts electrons and is reduced to water, and without it the whole chain backs up and oxidative phosphorylation halts. Water is the product of that reduction, not the acceptor, and NAD^+ and FAD are upstream carriers that deliver electrons into the chain rather than final acceptors.

13. The ATP synthase enzyme is located in the:

- A) Inner mitochondrial membrane
- B) Cytosol
- C) Outer mitochondrial membrane
- D) Thylakoid stroma
- E) Nuclear envelope

Answer: A

ATP synthase (complex V) sits in the inner mitochondrial membrane, folded into cristae, and uses the proton gradient between intermembrane space and matrix to drive ADP phosphorylation. The outer membrane and cytosol are not involved, and the thylakoid reference confuses chloroplast anatomy: the chloroplast ATP synthase spans the thylakoid membrane, not a stroma.

14. During which phase of cellular respiration is the majority of CO₂ released?

- A)** Oxidative phosphorylation
- B)** Pyruvate oxidation and the Krebs cycle
- C)** Lactic acid fermentation
- D)** Glycolysis
- E)** Substrate-level phosphorylation

Answer: B

Glycolysis and oxidative phosphorylation release no CO₂ at all. Pyruvate oxidation releases one CO₂ per pyruvate and each turn of the Krebs cycle releases two, so all six CO₂ molecules per glucose come from pyruvate oxidation together with the Krebs cycle, making that pair the sole source.

15. Which organelle is the primary site of protein synthesis for proteins destined for secretion?

- A)** Peroxisome
- B)** Nucleolus
- C)** Smooth endoplasmic reticulum
- D)** Lysosome
- E)** Rough endoplasmic reticulum

Answer: E

Secretory proteins are synthesised by ribosomes docked on the rough ER, whose studding by ribosomes gives it its name, and they then travel to the Golgi for processing. The smooth ER handles lipids and detoxification, and the nucleolus builds ribosomal subunits, not secreted proteins.

16. Which of the following is a function of the Golgi apparatus?

- A)** Storage of genetic information
- B)** Detoxification of hydrogen peroxide
- C)** ATP production via oxidative phosphorylation
- D)** Synthesis of ribosomal RNA
- E)** Modification, sorting, and packaging of proteins

Answer: E

The Golgi receives vesicles from the ER, chemically modifies its cargo (for example by glycosylation), sorts it, and packages it into vesicles for dispatch. The other listed jobs belong elsewhere: ribosomal RNA is made in the nucleolus, oxidative phosphorylation occurs in mitochondria, hydrogen peroxide breakdown is catalase in peroxisomes, and genetic information is stored in the nucleus.

17. Which structure is responsible for the selective permeability of the cell membrane?

- A)** The cytoskeleton
- B)** The nuclear pore complex alone
- C)** The cell wall of cellulose
- D)** The phospholipid bilayer with embedded proteins
- E)** The peptidoglycan layer

Answer: D

Selective permeability is a property of the phospholipid bilayer itself, whose hydrophobic core blocks ions and large polar molecules, while embedded channel and carrier proteins admit specific solutes. Cell walls such as peptidoglycan or cellulose are porous structural layers rather than selective barriers, and the cytoskeleton provides shape and transport tracks, not permeability.

18. Which transport mechanism requires the direct hydrolysis of ATP to move a substance against its concentration gradient?

- A)** Simple diffusion
- B)** Facilitated diffusion via channels
- C)** Primary active transport
- D)** Passive carrier-mediated diffusion
- E)** Osmosis

Answer: C

Primary active transport couples movement against the gradient directly to ATP hydrolysis, as in the sodium-potassium pump. Simple diffusion, osmosis, and channel- or carrier-mediated facilitated diffusion are all passive and run downhill on the gradient, so they consume no ATP at all.

19. Which of the following correctly describes the induced-fit model of enzyme action?

- A)** The substrate is permanently altered and the enzyme consumed.
- B)** The enzyme changes shape upon substrate binding to improve the fit.
- C)** The enzyme raises ΔG of the reaction.
- D)** Competitively bound inhibitors bind allosteric sites.
- E)** The active site is rigid and only one substrate can bind.

Answer: B

In the induced-fit model the active site is flexible and changes conformation when the substrate binds, moulding around it to align the catalytic groups. The rigid-site picture is the older lock-and-key model, and nothing in induced fit changes the free energy of the reaction or consumes the enzyme.

20. A competitive inhibitor of an enzyme:

- A)** Increases K_m but does not change V_{max} .
- B)** Decreases V_{max} but does not change K_m .
- C)** Binds covalently to the active site, inactivating the enzyme permanently.
- D)** Cannot be overcome by increasing substrate concentration.
- E)** Binds reversibly at a site other than the active site.

Answer: A

A competitive inhibitor binds reversibly at the active site and competes with the substrate, so more substrate is needed to reach half-maximal rate: the apparent K_m increases while V_{max} stays the same, because saturating substrate outcompetes the inhibitor. The pattern of a lowered V_{max} with unchanged K_m belongs to non-competitive inhibition instead, and no covalent permanent inactivation is involved.

21. Which type of muscle is involuntary, striated, and has branched cells connected by intercalated discs?

- A)** Myoepithelium
- B)** Smooth muscle
- C)** Cardiac muscle
- D)** Skeletal muscle
- E)** Tendons

Answer: C

Cardiac muscle is involuntary, striated and built of branched cells joined end to end by intercalated discs, whose gap junctions synchronise contraction. Skeletal muscle is striated but voluntary with unbranched fibres, and smooth muscle is involuntary but not striated, so only cardiac tissue satisfies all three descriptors at once.

22. The functional unit of contraction in skeletal muscle is the:

- A)** Sarcolemma
- B)** Sarcoplasm
- C)** Sarcoplasmic reticulum
- D)** Myofibril
- E)** Sarcomere

Answer: E

The sarcomere, the segment from one Z-disc to the next, is the smallest unit that shortens, so it is the functional unit of skeletal muscle. A myofibril is the whole chain of sarcomeres inside a fibre, the sarcolemma is the cell membrane, and the sarcoplasmic reticulum is the calcium store.

23. During the sliding filament mechanism, myosin binds to:

- A)** Myosin-binding site on titin
- B)** Actin
- C)** Troponin
- D)** Tropomyosin
- E)** Calcium directly

Answer: B

During contraction, calcium binds troponin, tropomyosin shifts away from the binding sites, and the myosin heads attach to actin to form cross-bridges that pull. Calcium regulates the process through troponin and does not bind myosin directly, and titin is the elastic scaffold rather than the myosin partner.

24. Oxygen is transported in blood primarily:

- A)** Within leucocytes
- B)** Bound to albumin
- C)** Dissolved in plasma
- D)** Bound to haemoglobin in red blood cells
- E)** As bicarbonate ions

Answer: D

About 98 percent of oxygen travels bound to haemoglobin inside red blood cells, with only a small remainder dissolved in plasma. Bicarbonate is the main transport form of carbon dioxide, not oxygen, and albumin or leucocytes play no role in oxygen carriage.

25. In which part of the human digestive system does the majority of nutrient absorption occur?

- A)** Small intestine
- B)** Large intestine
- C)** Stomach
- D)** Oesophagus
- E)** Liver

Answer: A

Most nutrient absorption occurs in the small intestine, whose villi and microvilli create an enormous surface area and whose length gives long contact time, with the jejunum the busiest stretch. The stomach mainly digests, the large intestine mostly reclaims water and ions, and the oesophagus and liver absorb essentially nothing.

26. Which hormone, produced by the beta cells of the pancreas, lowers blood glucose concentration?

- A)** Glucagon **B)** Adrenaline **C)** Thyroxine **D)** Insulin **E)** Cortisol

Answer: D

Beta cells of the pancreatic islets secrete insulin, which promotes glucose uptake into cells and storage as glycogen, lowering blood glucose. Glucagon, released by alpha cells, does the opposite and raises glucose, while cortisol, adrenaline and thyroxine also tend to raise blood glucose rather than lower it.

27. Which hormone is released from the posterior pituitary and causes water reabsorption in the kidney collecting duct?

- A)** Antidiuretic hormone (ADH / vasopressin)
B) Atrial natriuretic peptide
C) Growth hormone
D) Aldosterone
E) Prolactin

Answer: A

Antidiuretic hormone (ADH, vasopressin) is released from the posterior pituitary and causes the collecting duct to insert aquaporin channels, so water is reabsorbed and urine is concentrated. Aldosterone instead promotes sodium reabsorption from the adrenal cortex, and atrial natriuretic peptide promotes sodium loss, the reverse of ADH.

28. Which bone of the upper limb is the humerus? It is the:

- A)** Thigh bone
B) Upper arm bone
C) Shin bone
D) Forearm bone (thumb side)
E) Collar bone

Answer: B

The humerus is the single long bone of the upper arm, running from shoulder to elbow. The femur is the thigh bone, the radius is the thumb-side forearm bone, the tibia is the shin bone, and the clavicle is the collar bone, so only the upper-arm description matches.

29. In DNA replication, the enzyme that unwinds the double helix at the replication fork is:

- A)** Telomerase **B)** Ligase **C)** DNA polymerase **D)** Primase **E)** Helicase

Answer: E

Helicase unwinds the double helix by breaking the hydrogen bonds between base pairs at the replication fork. DNA polymerase then builds the new strands, primase lays the RNA primers, and ligase seals the nicks between Okazaki fragments, so the unwinding job belongs to helicase alone.

30. In a diploid organism with $2n = 46$, how many chromosomes are present in a mature gamete?

- A)** 23 **B)** 44 **C)** 46 **D)** 48 **E)** 92

Answer: A

A diploid count of $2n = 46$ means body cells carry 46 chromosomes; meiosis halves this, so a mature gamete carries the haploid number $n = 23$. Fertilisation then restores 46 in the zygote.

31. A man with blood group AB and a woman with blood group O have a child. Which blood groups are possible in their offspring?

- A)** A and AB
- B)** AB only
- C)** A and B only
- D)** O only
- E)** A, B, AB, and O

Answer: C

The AB father contributes either the A or the B allele, while the O mother can pass only the recessive allele i. Every child is therefore Ai (group A) or Bi (group B): AB is impossible because the mother supplies no A or B allele, and O is impossible because the father always supplies A or B. So only groups A and B can appear among the offspring.

32. During translation, the codon AUG serves to:

- A)** Bind release factors
- B)** Code for tryptophan only
- C)** Act as the start codon and code for methionine
- D)** Recruit RNA polymerase
- E)** Terminate translation

Answer: C

AUG is the start codon: it sets the reading frame and codes for methionine (N-formylmethionine in bacteria). Termination is handled by the stop codons UAA, UAG and UGA together with release factors, and RNA polymerase is recruited to DNA promoters, never by codons.

Section C - Chemistry (Q33-47)

33. What is the oxidation number of sulfur in H_2SO_4 ?

- A)** -2 **B)** 0 **C)** +2 **D)** +4 **E)** +6

Answer: E

Given: H_2SO_4 is a neutral molecule; standard values $\text{H} = +1$ and $\text{O} = -2$; let x be the oxidation number of S.

Relationship: oxidation numbers in a neutral molecule sum to 0.

Substitute: $2 \times (+1) + x + 4 \times (-2) = 0$

Hop: hydrogen gives $2 \times (+1) = +2$ and oxygen gives $4 \times (-2) = -8$, so $2 + x - 8 = 0$.

Hop: $x - 6 = 0$

Result: $x = +6$

Classic slip: +2 comes from taking each oxygen as -1 instead of -2, and +4 from dropping one oxygen from the count.

34. In the reaction $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$, which species is reduced?

- A)** Zn **B)** Cu^{2+} **C)** Zn^{2+} **D)** Cu (metal) **E)** SO_4^{2-}

Answer: B

In $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$ the copper ion gains two electrons ($\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$), so Cu^{2+} is reduced, while zinc loses electrons to become Zn^{2+} and is oxidised. The sulfate ion is a spectator throughout, and metallic copper is the reduced product rather than the species being reduced.

35. An acidic aqueous solution at 25 °C has a pH of 3. The concentration of H^+ is:

- A) $1 \times 10^{-3} \text{ M}$ B) $1 \times 10^3 \text{ M}$ C) $1 \times 10^{-11} \text{ M}$ D) $3 \times 10^{-1} \text{ M}$ E) 3 M

Answer: A

Given: $\text{pH} = 3$ at 25 °C; definition $\text{pH} = -\log_{10}[\text{H}^+]$

Relationship: inverting the logarithm gives $[\text{H}^+] = 10^{(-\text{pH})}$

Substitute: $[\text{H}^+] = 10^{(-3)} \text{ M}$

Hop: $10^{(-3)} = 1/1000 = 0.001$

Result: $[\text{H}^+] = 1 \times 10^{-3} \text{ M}$

Classic slip: $1 \times 10^{-11} \text{ M}$ is the hydroxide concentration ($\text{pOH} = 14 - 3 = 11$), the classic swap of H^+ for OH^- .

36. What is the pH of a 0.01 M solution of a strong monoprotic acid (e.g., HCl) at 25 °C?

- A) 1 B) 2 C) 3 D) 12 E) 13

Answer: B

Given: strong monoprotic acid HCl at $c = 0.01 \text{ M}$; a strong acid dissociates completely and releases one H^+ per molecule.

Relationship: $[\text{H}^+] = c$, and $\text{pH} = -\log_{10}[\text{H}^+]$

Substitute: $[\text{H}^+] = 0.01 \text{ M} = 10^{-2} \text{ M}$, so $\text{pH} = -\log_{10}(10^{-2})$

Hop: $-\log_{10}(10^{-2}) = -(-2) = 2$

Result: $\text{pH} = 2$

Classic slip: 12 or 13 runs the numbers as if for pOH (subtracting from 14); for an acid you take the negative logarithm of the hydrogen ion concentration directly.

37. Which type of bond involves the complete transfer of electrons from one atom to another?

- A) Metallic bond
B) Van der Waals interaction
C) Hydrogen bond
D) Ionic bond
E) Covalent bond

Answer: D

An ionic bond forms by complete transfer of electrons from one atom to another, producing oppositely charged ions held together by electrostatic attraction, as when sodium donates an electron to chlorine. Covalent bonds share electrons, hydrogen bonds and van der Waals interactions are weak intermolecular attractions, and metallic bonding uses a delocalised electron sea.

38. How many electrons are shared in a double covalent bond between two atoms?

- A) 1 B) 2 C) 3 D) 4 E) 6

Answer: D

Given: a double covalent bond consists of two shared electron pairs.

Relationship: each shared pair holds 2 electrons, so electrons shared = pairs $\times$ 2.

Substitute: 2 pairs $\times$ 2 electrons per pair.

Hop: $2 \times 2 = 4$

Result: 4 electrons are shared, as in the $\text{O}=\text{O}$ double bond.

Classic slip: 2 counts a single shared pair, which is the single-bond case.

39. At constant temperature and pressure, the volume of a gas is directly proportional to the number of moles. This is:

- A)** Boyle's law **B)** Dalton's law **C)** Avogadro's law **D)** Charles's law **E)** Henry's law

Answer: C

Volume proportional to the number of moles at constant temperature and pressure is Avogadro's law ($V \propto n$). Boyle's law covers $V \propto 1/P$ at fixed n and T , and Charles's law covers $V \propto T$ at fixed n and P , so the mole-proportionality is the distinctive signature of Avogadro's law.

40. Calculate the number of molecules in 0.5 mol of N_2 (Avogadro's number = 6.022×10^{23}).

- A)** 1.20×10^{24} **B)** 1.50×10^{23} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 6.02×10^{22}

Answer: C

Given: $n = 0.5$ mol of N_2 ; Avogadro's number = 6.022×10^{23} molecules per mol

Relationship: number of molecules = moles $\times$ Avogadro's number.

Substitute: $N = 0.5 \text{ mol} \times 6.022 \times 10^{23} \text{ molecules/mol}$

Hop: $6.022 \times 10^{23} \times 0.5 = 3.011 \times 10^{23}$

Result: $N \approx 3.01 \times 10^{23}$ molecules.

Classic slip: 6.02×10^{23} forgets the 0.5 and answers for a full mole instead of half a mole.

41. Which element is in Group 17 and Period 3 of the periodic table?

- A)** Fluorine **B)** Iodine **C)** Sulfur **D)** Bromine **E)** Chlorine

Answer: E

Period 3 runs Na, Mg, Al, Si, P, S, Cl, Ar, and the Group 17 halogens fall F (period 2), Cl (period 3), Br (period 4), I (period 5). The intersection is chlorine. Fluorine has the right group but sits in period 2, and sulfur is period 3 but Group 16.

42. Which subatomic particle has no electric charge and a mass of approximately 1 amu?

- A)** Neutron **B)** Proton **C)** Photon **D)** Positron **E)** Electron

Answer: A

The neutron carries no electric charge and has a mass of about 1 amu, essentially equal to the proton.

The proton is positively charged, the electron is negative and roughly $1/1836$ of an amu, and a photon is a massless quantum of electromagnetic radiation.

43. For the equilibrium $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, $\Delta H < 0$. Increasing the pressure at constant temperature will:

- A)** Shift the equilibrium to the left, producing more N_2 and H_2 .
- B)** Shift equilibrium to the right, producing more NH_3
- C)** Decrease the rate only
- D)** Increase the temperature
- E)** Have no effect

Answer: B

Givens: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$; pressure increased at constant temperature.

Deduction: count gas moles on each side: left = $1 + 3 = 4$ mol; right = 2 mol.

Deduction: by Le Chatelier's principle, raising the pressure shifts the equilibrium toward the side with fewer gas moles.

Deduction: 2 mol is fewer than 4 mol, so the balance moves to the right.

Conclusion: more NH_3 is produced (the exothermic ΔH plays no role, since temperature is unchanged).

Tempting claim: that nothing changes confuses the equilibrium position with the rate; the shift happens precisely because the two sides hold different mole counts.

44. The functional group $-\text{COOH}$ characterises which class of organic compounds?

- A)** Alcohols
- B)** Ethers
- C)** Aldehydes
- D)** Carboxylic acids
- E)** Ketones

Answer: D

The COOH (carboxyl) group is the signature of carboxylic acids. Alcohols carry OH , aldehydes carry CHO , ketones have an internal $\text{C}=\text{O}$, and ethers have a $\text{C}-\text{O}-\text{C}$ linkage, so the acidic carboxyl description decides the question.

45. Which of the following compounds is an alkane?

- A)** C_2H_4 **B)** C_2H_2 **C)** C_6H_6 **D)** C_3H_6 **E)** C_4H_{10}

Answer: E

Given: alkanes follow the general formula $\text{C}_n\text{H}_{2n+2}$; candidates are C_2H_4 , C_3H_6 , C_4H_{10} , C_6H_6 and C_2H_2 .

Relationship: a formula is an alkane only if its hydrogen count equals $2 \times (\text{carbon count}) + 2$.

Hops: C_2H_4 : $2 \times 2 + 2 = 6$, not 4; C_3H_6 : $2 \times 3 + 2 = 8$, not 6; C_4H_{10} : $2 \times 4 + 2 = 10$, an exact match.

Result: C_4H_{10} (butane) is the alkane.

Remaining check: C_2H_2 fits $\text{C}_n\text{H}_{2n-2}$ (an alkyne) and C_6H_6 is aromatic benzene, so neither qualifies.

Classic slip: accepting C_3H_6 as an alkane by eye; it satisfies C_nH_{2n} , the alkene or cycloalkane family, not the alkane rule.

46. The rate of a chemical reaction generally increases with temperature because:

- A)** The activation energy decreases.
- B)** The reaction becomes exothermic.
- C)** The equilibrium constant always increases.
- D)** Catalysts form spontaneously.
- E)** More collisions occur with energy $\geq E_a$ per unit time.

Answer: E

Raising the temperature increases the fraction of colliding molecules whose energy meets or exceeds the activation energy (the Arrhenius picture, $k = A \cdot e^{(-E_a/RT)}$) and also raises collision frequency, so the rate rises. The activation energy itself is fixed by the reaction pathway and does not decrease with heating, and the equilibrium constant grows with temperature only for endothermic reactions, so it does not always increase.

47. What is the molar concentration of a solution containing 0.2 mol of NaCl dissolved to make 500 mL of solution?

- A)** 0.1 M **B)** 0.2 M **C)** 0.4 M **D)** 0.5 M **E)** 1.0 M

Answer: C

Given: moles of NaCl = 0.2 mol; final solution volume = 500 mL.

Relationship: molarity c = moles of solute $\div$ volume of solution in litres.

Convert: 500 mL = $500 \div 1000 = 0.500$ L

Substitute: $c = 0.2 \text{ mol} \div 0.500 \text{ L}$

Hop: $0.2 \div 0.5 = 0.4$

Result: $c = 0.4 \text{ M}$

Classic slip: 0.2 M quotes the mole amount without ever dividing by the volume; the litres conversion must come first.

Section D - Mathematics (Q48-54)

48. Solve for x : $\sqrt{2x - 1} = 3$.

- A)** $x = 2$ **B)** $x = 3$ **C)** $x = 6$ **D)** $x = 5$ **E)** $x = 4$

Answer: D

Given: $\sqrt{2x - 1} = 3$; the expression under the root must be ≥ 0 .

Relationship: squaring both sides removes the root, since $(\sqrt{a})^2 = a$.

Substitute: $(\sqrt{2x - 1})^2 = 3^2$ gives $2x - 1 = 9$

Hop: $2x = 9 + 1 = 10$

Hop: $x = 10 \div 2 = 5$

Check: $\sqrt{2 \times 5 - 1} = \sqrt{9} = 3$, as required

Result: $x = 5$

Classic slip: an unchecked value such as $x = 4$ leaves $\sqrt{7}$ on the left instead of 3; always substitute back into the original equation.

49. Solve: $2x - 5 < 3x + 1$.

- A) $x > -6$ B) $x < 6$ C) $x < -6$ D) $x = -6$ E) $x > 6$

Answer: A

Given: $2x - 5 < 3x + 1$.

Relationship: collect x on one side and numbers on the other; the sign flips only when both sides are divided by a negative number.

Hop: subtract $2x$ from both sides: $-5 < x + 1$.

Hop: subtract 1 from both sides: $-6 < x$.

Result: $-6 < x$, written with x first as $x > -6$.

Classic slip: $x < -6$ flips the inequality for no reason, as if a negative divisor had appeared; no division occurred, so the direction never changes.

50. In a triangle, two angles are 50° and 60° . What is the third angle?

- A) 50° B) 60° C) 70° D) 80° E) 90°

Answer: C

Given: two angles of the triangle, 50° and 60° .

Relationship: interior angles of a triangle sum to 180° , so the third angle = $180^\circ - 50^\circ - 60^\circ$.

Substitute: third = $180^\circ - 50^\circ - 60^\circ$

Hop: $50^\circ + 60^\circ = 110^\circ$

Hop: $180^\circ - 110^\circ = 70^\circ$

Result: the third angle is 70° .

Classic slip: 80° comes from adding the two given angles as 100° instead of 110° before subtracting from 180° .

51. If $f(x) = 2x^2 - 3x + 1$, what is $f(2)$?

- A) 1 B) 2 C) 3 D) 5 E) 7

Answer: C

Given: $f(x) = 2x^2 - 3x + 1$; evaluate at $x = 2$

Relationship: replace every x by 2: $f(2) = 2 \times (2)^2 - 3 \times 2 + 1$

Hop: $(2)^2 = 4$, so the first term is $2 \times 4 = 8$

Hop: the second term is $3 \times 2 = 6$

Hop: $8 - 6 + 1 = 2 + 1 = 3$

Result: $f(2) = 3$

Classic slip: 2 forgets the constant +1 and stops at $8 - 6$; carry every term of the function.

52. A circle has radius 5 cm. What is its area? (Take $\pi = 3.14$.)

- A) 15.7 cm² B) 25 cm² C) 31.4 cm² D) 62.8 cm² E) 78.5 cm²

Answer: E

Given: radius $r = 5$ cm; $\pi = 3.14$

Relationship: area of a circle $A = \pi r^2$

Substitute: $A = 3.14 \times (5 \text{ cm})^2$

Hop: $(5 \text{ cm})^2 = 25 \text{ cm}^2$

Hop: $3.14 \times 25 = 78.5$

Result: $A = 78.5 \text{ cm}^2$

Classic slip: 31.4 cm² is the circumference $2 \times 3.14 \times 5$ wearing area units, from never squaring the radius.

53. A bag contains 3 red and 2 blue balls. One ball is drawn at random. What is the probability it is blue?

- A) 1/5 B) 1/2 C) 2/5 D) 2/3 E) 3/5

Answer: C

Given: 3 red balls and 2 blue balls; one ball drawn at random.

Relationship: probability = favourable outcomes $\div$ total outcomes.

Substitute: total = $3 + 2 = 5$; favourable (blue) = 2

Hop: $P(\text{blue}) = 2 \div 5 = 2/5$

Result: the probability of drawing a blue ball is 2/5.

Classic slip: 3/5 is the complementary event, drawing red, from placing the red count on top.

54. A rectangle has length 12 cm and diagonal 13 cm. What is its perimeter?

- A) 24 cm B) 28 cm C) 30 cm D) 34 cm E) 36 cm

Answer: D

Given: length $l = 12$ cm; diagonal $d = 13$ cm; the diagonal, the length and the width form a right triangle.

Relationship: Pythagoras, $d^2 = l^2 + w^2$, so $w = \sqrt{(d^2 - l^2)}$

Substitute: $w = \sqrt{(13^2 - 12^2)} \text{ cm}$

Hops: $13^2 = 169$; $12^2 = 144$; $169 - 144 = 25$; $\sqrt{25} = 5$, so $w = 5$ cm

Relationship: perimeter = $2 \times (l + w)$

Substitute: $2 \times (12 + 5) \text{ cm} = 2 \times 17 \text{ cm}$

Hop: $2 \times 17 = 34$

Result: the perimeter is 34 cm.

Classic slip: 24 cm doubles only the length and never finds the width; both dimensions are needed.

Section E - Physics (Q55-60)

55. A car travels 120 km in 2 hours at constant velocity. What is its speed in m/s?

- A)** 16.7 m/s **B)** 33.3 m/s **C)** 50 m/s **D)** 60 m/s **E)** 120 m/s

Answer: A

Given: distance $s = 120$ km; time $t = 2$ h; constant velocity.

Relationship: speed $v = s \div t$; convert with $1 \text{ km} = 1000 \text{ m}$ and $1 \text{ h} = 3600 \text{ s}$

Substitute: $v = 120 \text{ km} \div 2 \text{ h} = 60 \text{ km/h}$

Convert: $60 \text{ km/h} = 60 \times 1000 \text{ m} \div 3600 \text{ s} = 60\,000 \text{ m} \div 3600 \text{ s}$

Hop: $60\,000 \div 3600 = 16.666... \approx 16.7$

Result: $v \approx 16.7 \text{ m/s}$.

Classic slip: 60 m/s just relabels the km/h figure, forgetting that 1 km/h is only $1000/3600 \approx 0.28 \text{ m/s}$.

56. Three resistors of 2Ω , 3Ω , and 6Ω are connected in parallel. What is the equivalent resistance?

- A)** 0.5Ω **B)** 1Ω **C)** 3.66Ω **D)** 6Ω **E)** 11Ω

Answer: B

Given: three resistors in parallel, 2Ω , 3Ω and 6Ω .

Relationship: for a parallel combination, $1/R = 1/R_1 + 1/R_2 + 1/R_3$

Substitute: $1/R = 1/2 + 1/3 + 1/6$ (in $1/\Omega$)

Hop: with common denominator 6: $3/6 + 2/6 + 1/6 = 6/6 = 1$

Hop: invert: $R = 1 \div 1 = 1 \Omega$

Result: $R = 1 \Omega$

Classic slip: 11Ω adds the resistances, which is the series rule; a parallel equivalent must come out smaller than the smallest branch (here smaller than 2Ω).

57. A current of 2 A flows through a 5Ω resistor. What is the power dissipated?

- A)** 5 W **B)** 10 W **C)** 20 W **D)** 50 W **E)** 100 W

Answer: C

Given: current $I = 2 \text{ A}$; resistance $R = 5 \Omega$

Relationship: power dissipated in a resistor $P = I^2 R$

Substitute: $P = (2 \text{ A})^2 \times 5 \Omega$

Hop: $(2 \text{ A})^2 = 4 \text{ A}^2$

Hop: $4 \times 5 = 20$

Result: $P = 20 \text{ W}$

Classic slip: 10 W stops at the voltage drop $V = IR = 2 \times 5 = 10 \text{ V}$ and calls it watts; the power needs the extra factor of current.

58. A book rests on a table. The reaction force of the table on the book is an example of which law?

- A)** Hooke's law
- B)** Newton's third law
- C)** Newton's second law only
- D)** Newton's first law only
- E)** Ohm's law

Answer: B

Givens: a book rests on a table; the force asked about is the table's push up on the book.

Deduction: Newton's third law pairs two forces that act on different bodies, equal in size and opposite in direction.

Deduction: the upward push of table on book is paired with the equal downward push of book on table.

Conclusion: this reaction force is an example of Newton's third law.

Tempting claim: the first law only explains why the book stays at rest under zero net force; it pairs no forces, and the book's weight pairs with the Earth, not with the table's push.

59. The gravitational force between two bodies is $F = Gm_1m_2/r^2$. If the distance between the two bodies is halved, the gravitational force becomes:

- A)** $2F$
- B)** F
- C)** $F/2$
- D)** $4F$
- E)** $F/4$

Answer: D

Given: $F = Gm_1m_2/r^2$; new distance $r/2$; masses unchanged.

Relationship: substitute the new distance into the inverse-square law.

Substitute: $F' = Gm_1m_2 \div (r/2)^2$

Hop: $(r/2)^2 = r^2/4$, so $F' = Gm_1m_2 \div (r^2/4) = 4 \times Gm_1m_2/r^2$

Result: $F' = 4F$

Classic slip: $2F$ inverts the distance only once (multiplying by 2) instead of squaring the inversion (multiplying by 4); the square in the denominator applies to the halved distance as well.

60. The pressure at depth h in a fluid of density ρ is given by $P = \rho gh$. If $\rho = 1000 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$, and $h = 10 \text{ m}$, what is the gauge pressure?

- A)** 9,800 Pa
- B)** 98 Pa
- C)** 98,000 Pa
- D)** 980 Pa
- E)** 980,000 Pa

Answer: C

Given: $\rho = 1000 \text{ kg/m}^3$; $g = 9.8 \text{ m/s}^2$; depth $h = 10 \text{ m}$; gauge pressure asked.

Relationship: gauge pressure $P = \rho gh$; gauge values exclude atmospheric pressure, so nothing is added.

Substitute: $P = 1000 \text{ kg/m}^3 \times 9.8 \text{ m/s}^2 \times 10 \text{ m}$

Hop: $1000 \times 9.8 = 9800$

Hop: $9800 \times 10 = 98\,000$

Result: $P = 98\,000 \text{ Pa}$

Classic slip: 980 000 Pa slips in an extra factor of ten, and 980 Pa is two powers of ten low; track the magnitude of all three factors.